

Chapter 8.

Advanced Counting Techniques

進階計數技巧

- 8.1 Applications of Recurrence Relations**
- 8.2 Solving Linear Recurrence Relations**
- 8.3 Divide-and-Conquer Algorithms and Recurrence Relations**
- 8.4 Generating Functions** 生成函數
- 8.5 Inclusion-Exclusion**
- 8.6 Applications of Inclusion-Exclusion**

Recurrence Relations

(recalling definitions from Chapter 2)

Definition: A **recurrence relation** (遞迴關係) for the sequence $\{a_n\}$ is an equation that expresses a_n in terms of one or more of the previous terms of the sequence, namely, $a_0, a_1, \dots, a_{n-1}$, for all integers n with $n \geq n_0$, where n_0 is a nonnegative integer.

- A sequence is called a **solution** of a recurrence relation if its terms satisfy the recurrence relation.
- The **initial conditions** for a sequence specify the terms that precede the first term where the recurrence relation takes effect.

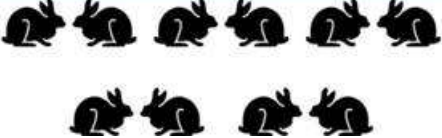
Rabbits and the Fibonacci Numbers₁

*費式數列

Example: A young pair of rabbits (one of each gender) is placed on an island. A pair of rabbits does not breed until they are 2 months old. **After they are 2 months old, each pair of rabbits produces another pair each month.** Find a recurrence relation for the number of pairs of rabbits on the island after n months, assuming that rabbits never die. $f(n) = f(n-1) + f(n-2)$

*This is the original problem considered by **Leonardo Pisano (Fibonacci)** in the thirteenth century.*

Rabbits and the Fibonacci Numbers₂

Reproducing pairs (at least two months old)	Young pairs (less than two months old)	Month	Reproducing pairs	Young pairs	Total pairs
		1	0	1	1
		2	0	1	1
		3	1	1	2
		4	1	2	3
		5	2	3	5
		6	3	5	8

Modeling the Population Growth of Rabbits on an Island

[Jump to long description](#)

Rabbits and the Fibonacci Numbers₃

Solution: Let f_n be the number of pairs of rabbits after n months.

- There are $f_1 = 1$ pairs of rabbits on the island at the end of the first month.
- We also have $f_2 = 1$ because the pair does not breed during the first month.
- To find the number of pairs on the island after n months, add the number on the island after the previous month, f_{n-1} , and the number of newborn pairs, which equals f_{n-2} , because each newborn pair comes from a pair at least two months old.

Consequently the sequence $\{f_n\}$ satisfies the recurrence relation

$$f_n = f_{n-1} + f_{n-2} \text{ for } n \geq 3 \text{ with the initial conditions } f_1 = 1 \text{ and } f_2 = 1.$$

The number of pairs of rabbits on the island after n months is given by the n th Fibonacci number.

The Tower of Hanoi₁ 問題:

(河內塔)

① 幾步

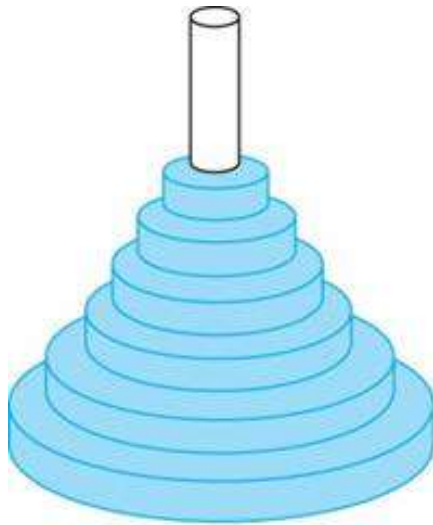
② 方法

In the late nineteenth century, the French mathematician Édouard Lucas invented a puzzle consisting of three pegs on a board with disks of different sizes. Initially all of the disks are on the first peg in order of size, with the largest on the bottom.

Rules: You are allowed to move the disks one at a time from one peg to another as long as **a larger disk is never placed on a smaller.**

Goal: Using allowable moves, end up with all the disks on the second peg in order of size with largest on the bottom.

The Tower of Hanoi₂



Peg 1



Peg 2

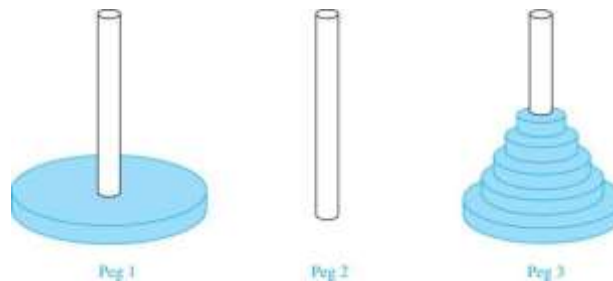


Peg 3

The Initial Position in the Tower of Hanoi Puzzle

The Tower of Hanoi₃

Solution: Let $\{H_n\}$ denote the number of moves needed to solve the Tower of Hanoi Puzzle with n disks. Set up a recurrence relation for the sequence $\{H_n\}$. Begin with n disks on peg 1. We can transfer the top $n - 1$ disks, following the rules of the puzzle, to peg 3 using H_{n-1} moves.

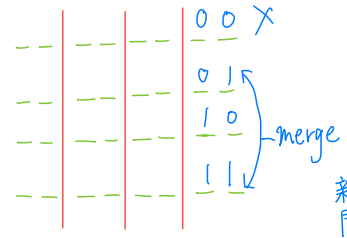


First, we use 1 move to transfer the largest disk to the second peg. Then we transfer the $n - 1$ disks from peg 3 to peg 2 using H_{n-1} additional moves. This can not be done in fewer steps. Hence,

$$H_n = 2H_{n-1} + 1.$$

The initial condition is $H_1 = 1$ since a single disk can be transferred from peg 1 to peg 2 in one move.

長度 n :



Counting Bit Strings₁

→ 結尾是 1, 或結尾是 0。→ 然後分解成 $n-1$ 項, 繼續解下去

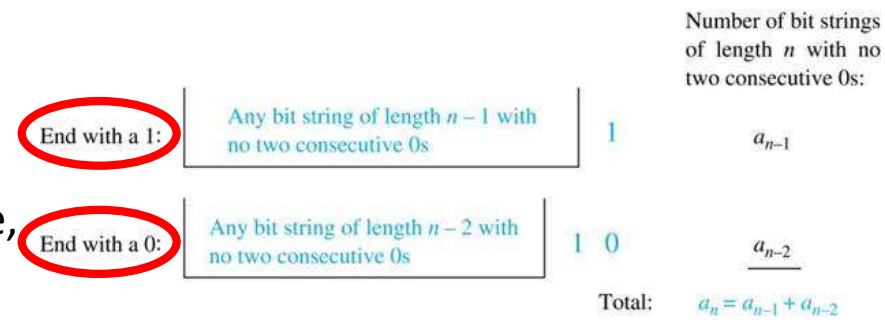
Example 3: Find a recurrence relation and give initial conditions for **the number of bit strings of length n without two consecutive 0s**. How many such bit strings are there of length five?

不能有兩個連續 0

Solution: Let a_n denote the number of bit strings of length n without two consecutive 0s. To obtain a recurrence relation for $\{a_n\}$ note that the number of bit strings of length n that do not have two consecutive 0s is the number of bit strings ending with a 0 plus the number of such bit strings ending with a 1.

Now assume that $n \geq 3$.

- The bit strings of length n ending with 1 without two consecutive 0s are the bit strings of length $n - 1$ with no two consecutive 0s with a 1 at the end. Hence, there are a_{n-1} such bit strings.
- The bit strings of length n ending with 0 without two consecutive 0s are the bit strings of length $n - 2$ with no two consecutive 0s with 10 at the end. Hence, there are a_{n-2} such bit strings.



We conclude that $a_n = a_{n-1} + a_{n-2}$ for $n \geq 3$.

Bit Strings₂

The initial conditions are:

- $a_1 = 2$, since both the bit strings **0** and **1** do not have consecutive 0s.
- $a_2 = 3$, since the bit strings **01**, **10**, and **11** do not have consecutive 0s, while 00 does.

To obtain a_5 , we use the recurrence relation three times to find that:

- $a_3 = a_2 + a_1 = 3 + 2 = 5$
- $a_4 = a_3 + a_2 = 5 + 3 = 8$
- $a_5 = a_4 + a_3 = 8 + 5 = 13$

Note that $\{a_n\}$ satisfies the same recurrence relation as the Fibonacci sequence. Since $a_1 = f_3$ and $a_2 = f_4$, we conclude that $a_n = f_{n+2}$.

Counting the Ways to Parenthesize a Product

Example: Find a recurrence relation for C_n , the number of ways to parenthesize the product of $n + 1$ numbers, $x_0 \cdot x_1 \cdot x_2 \cdot \cdots \cdot x_n$, to specify the order of multiplication. For example, $C_3 = 5$, since all the possible ways to parenthesize 4 numbers are

$$((x_0 \cdot x_1) \cdot x_2) \cdot x_3 \qquad (x_0 \cdot (x_1 \cdot x_2)) \cdot x_3 \qquad (x_0 \cdot x_1) \cdot (x_2 \cdot x_3)$$

<先把最外层括号还他> $x_0 \cdot ((x_1 \cdot x_2) \cdot x_3)$ $x_0 \cdot (x_1 \cdot (x_2 \cdot x_3))$

Solution: Note that however parentheses are inserted in $x_0 \cdot x_1 \cdot x_2 \cdot \cdots \cdot x_n$, one “.” operator remains outside all parentheses. This final operator appears between two of the $n + 1$ numbers, say x_k and x_{k+1} . Since there are C_k ways to insert parentheses in the product $x_0 \cdot x_1 \cdot x_2 \cdot \cdots \cdot x_k$ and C_{n-k-1} ways to insert parentheses in the product $x_{k+1} \cdot x_{k+2} \cdot \cdots \cdot x_n$, we have

$$C_n = C_0 C_{n-1} + C_1 C_{n-2} + \cdots + C_{n-2} C_1 + C_{n-1} C_0$$
$$= \sum_{k=0}^{n-1} C_k C_{n-k-1}$$

$C_2 = C_0 C_1 + C_1 C_0$

<key>

$$(x_0) \times (x_1 \times x_2 \times x_3 \times x_4 \times x_5)$$
$$(x_0 \times x_1) \times (x_2 \times x_3 \times x_4 \times x_5)$$

$\vdots$

$$C_5 = C_0 \times C_4 + C_1 \times C_3 + C_2 \times C_2 + C_3 \times C_1 + C_4 \times C_0$$

The initial conditions are $C_0 = 1$ and $C_1 = 1$.

卡特兰数

The sequence $\{C_n\}$ is the sequence of **Catalan Numbers**. This recurrence relation can be solved using the method of generating functions; see Exercise 41 in Section 8.4.

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Linear Homogeneous Recurrence Relations

線性 齊次 (同次) 遞迴

Definition: A **linear homogeneous recurrence relation** of degree k with **constant** coefficients is a recurrence relation of the form $a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k}$, where $c_1, c_2, \dots, c_k$ are **real numbers**, and $c_k \neq 0$.

- it is **linear** because the right-hand side is a sum of the previous terms of the sequence each multiplied by a function of n .
- it is **homogeneous** (齊次) because **no terms occur that are not multiples of the a_j s**. Each coefficient is a constant.
- the **degree** is k because a_n is expressed in terms of the **previous k terms** of the sequence.

By strong induction, a sequence satisfying such a recurrence relation is uniquely determined by the recurrence relation and the k initial conditions $a_0 = C_1, a_1 = C_1, \dots, a_{k-1} = C_{k-1}$.

Examples of Linear Homogeneous Recurrence Relations

$P_n = (1.11) P_{n-1}$
✓ 是

linear homogeneous recurrence relation of degree one

$f_n = |f_{n-1} + |f_{n-2}$
✓ 都是1倍, 是

linear homogeneous recurrence relation of degree two

$a_n = a_{n-1} + (a_{n-2})^{(2)}$
每項都只能乘一個係數

not linear

$H_n = 2H_{n-1} + \underbrace{1}$
齊次, 不能多加東西

not homogeneous

$B_n = \underbrace{n} B_{n-1}$
h must a constant.

coefficients are not constants

Solving Linear Homogeneous Recurrence Relations

<解的基本結構>
: r^n , r must a constant

The basic approach is to look for solutions of the form $a_n = r^n$, where r is a constant. <e.g. 費式數列>
 $f_n = f_{n-1} + f_{n-2}$

Note that $a_n = r^n$ is a solution to the recurrence relation ↓
表示為 r^n

$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k}$ if and only if

$$r^n = c_1 r^{n-1} + c_2 r^{n-2} + \dots + c_k r^{n-k} \cdot \text{特徵式}$$

$r^n = r^{n-1} + r^{n-2}$
↓ 約掉 r^{n-2}
 $r^2 = r + 1 \rightarrow$ 求 r 的 r^2

Algebraic manipulation yields the **characteristic equation**:

$$r^k - c_1 r^{k-1} - c_2 r^{k-2} - \dots - c_{k-1} r - c_k = 0$$

The sequence $\{a_n\}$ with $a_n = r^n$ is a solution **if and only if** r is a **solution** to the characteristic equation.

The solutions to the characteristic equation are called the **characteristic roots** of the recurrence relation. The roots are used to give an explicit formula for all the solutions of the recurrence relation. 特徵根

Solving Linear Homogeneous Recurrence Relations of Degree Two

Theorem 1: Let c_1 and c_2 be real numbers. Suppose that $r^2 - c_1r - c_2 = 0$ has two distinct roots r_1 and r_2 . Then the sequence $\{a_n\}$ is a solution to the recurrence relation

$$a_n = c_1 a_{n-1} + c_2 a_{n-2}$$

假设该式有
两相异的根
(这还没有说根要是实数!)

if and only if

$$a_n = \alpha_1 r_1^n + \alpha_2 r_2^n$$

r_1^n 和 r_2^n 的线性组合

for $n = 0, 1, 2, \dots$, where α_1 and α_2 are constants.

可能不是實數根的 Case

$$r^2 + r + 1 = 0 \quad r = \frac{-1 \pm \sqrt{1-4}}{2}$$

$$r^2 = -r - 1 \quad = \frac{-1 \pm \sqrt{3}}{2}$$

$$r^n = -r^{n-1} - r^{n-2} = \frac{-1 \pm \sqrt{3}i}{2}$$

$$f_n = -f_{n-1} - f_{n-2}$$

$$f_n = \alpha_1 \left(\frac{-1 + \sqrt{3}i}{2} \right)^n + \alpha_2 \left(\frac{-1 - \sqrt{3}i}{2} \right)^n$$

$$f_0 = 1, f_1 = 2, f_2 = -3, f_3 = \dots$$

So: 重點不是在根是否實數,
而是根是否為 兩相異!

Using Theorem 1

Example: What is the solution to the recurrence relation

$$(r^n = r^{n-1}r + 2r^{n-2}) \Rightarrow (r^2 = r + 2) \quad \text{好耶!} \\ a_n = a_{n-1} + 2a_{n-2} \text{ with } a_0 = 2 \text{ and } a_1 = 7? \quad \downarrow \text{十字交乘}$$

Solution: The characteristic equation is $r^2 - r - 2 = 0$.
 $(r-2)(r+1) = 0$

Its roots are $r = 2$ and $r = -1$. Therefore, $\{a_n\}$ is a solution to the recurrence relation if and

only if $a_n = \alpha_1 2^n + \alpha_2 (-1)^n$, for some constants α_1 and α_2 .
 求出 r, 列出式!

To find the constants α_1 and α_2 , note that

$$a_0 = 2 = \alpha_1 + \alpha_2 \text{ and } a_1 = 7 = \alpha_1 2 + \alpha_2 (-1).$$

Solving these equations, we find that $\alpha_1 = 3$ and $\alpha_2 = -1$.

Hence, the solution is the sequence $\{a_n\}$ with $a_n = 3 \cdot 2^n - (-1)^n$.

然後, 要記得驗算!

18 An Explicit Formula for the Fibonacci Numbers₁

We can use Theorem 1 to find an explicit formula for the Fibonacci numbers. The sequence of Fibonacci numbers satisfies the recurrence relation $f_n = f_{n-1} + f_{n-2}$ with the initial conditions: $f_0 = 0$ and $f_1 = 1$.

Solution: The roots of the characteristic equation $r^2 - r - 1 = 0$ are

<Review 公式解>:

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$r_1 = \frac{1 + \sqrt{5}}{2}$$

$$r_2 = \frac{1 - \sqrt{5}}{2}$$

$$f_n = f_{n-1} + f_{n-2}$$

$$r^n = r^{n-1} + r^{n-2}$$

$$r^2 = r + 1$$

↓
×公式解



Fibonacci Numbers₂

Therefore by Theorem 1

$$f_n = \alpha_1 \left(\frac{1+\sqrt{5}}{2} \right)^n + \alpha_2 \left(\frac{1-\sqrt{5}}{2} \right)^n$$

for some constants α_1 and α_2 .

帶入初值求 α_1, α_2

Using the initial conditions $f_0 = 0$ and $f_1 = 1$, we have

$$f_0 = \alpha_1 + \alpha_2 = 0$$

$$f_1 = \alpha_1 \left(\frac{1+\sqrt{5}}{2} \right) + \alpha_2 \left(\frac{1-\sqrt{5}}{2} \right) = 1.$$

Solving, we obtain

$$\alpha_1 = \frac{1}{\sqrt{5}}, \quad \alpha_2 = -\frac{1}{\sqrt{5}}.$$

Hence,

$$f_n = \frac{1}{\sqrt{5}} \left(\frac{1+\sqrt{5}}{2} \right)^n - \frac{1}{\sqrt{5}} \left(\frac{1-\sqrt{5}}{2} \right)^n.$$

→ 然後帶入 $f_0, f_1, \dots$
多帶幾項驗算

²⁰The Solution when there is a Repeated Root

Theorem 2: Let c_1 and c_2 be real numbers with $c_2 \neq 0$. Suppose that $r^2 - c_1r - c_2 = 0$ has one **repeated root** r_0 . Then the sequence $\{a_n\}$ is a solution to the recurrence relation

$$\underline{a_n = c_1 a_{n-1} + c_2 a_{n-2}}$$

if and only if

$$a_n = \alpha_1 r_0^n + \alpha_2 n r_0^n$$

for $n = 0, 1, 2, \dots$, where α_1 and α_2 are constants.

重根, 多一个 nr^n
 $\langle r^n, nr^n \rangle$

重根

Using Theorem 2

Example: What is the solution to the recurrence relation

$$a_n = 6a_{n-1} - 9a_{n-2} \text{ with } a_0 = 1 \text{ and } a_1 = 6?$$

Handwritten notes: $r^n = 6r^{n-1} - 9r^{n-2} \Rightarrow r^2 = 6r - 9$ and $(r-3)^2$

Solution: The characteristic equation is $r^2 - 6r + 9 = 0$.

The only root is $r = 3$. Therefore, $\{a_n\}$ is a solution to the recurrence relation if and only if

$$a_n = \alpha_1 3^n + \alpha_2 n(3)^n$$

where α_1 and α_2 are constants.

To find the constants α_1 and α_2 , note that

$$a_0 = 1 = \alpha_1 \quad \text{and} \quad a_1 = 6 = \alpha_1 \cdot 3 + \alpha_2 \cdot 3.$$

Solving, we find that $\alpha_1 = 1$ and $\alpha_2 = 1$.

Hence,

$$a_n = 3^n + n3^n.$$

(r-3)^n \rightarrow \text{root } 3

為配合2個初值，一定要多生一個 $n3^n$ 在 α_2 項

22 Solving Linear Homogeneous Recurrence Relations of Arbitrary Degree

This theorem can be used to solve linear homogeneous recurrence relations with constant coefficients of any degree when the characteristic equation has distinct roots.

Theorem 3: Let $c_1, c_2, \dots, c_k$ be real numbers. Suppose that the characteristic equation

$$r^k - c_1 r^{k-1} - \dots - c_k = 0$$

has **k distinct roots** $r_1, r_2, \dots, r_k$. Then a sequence $\{a_n\}$ is a solution of the recurrence relation

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k}$$

if and only if

$$a_n = \alpha_1 r_1^n + \alpha_2 r_2^n + \dots + \alpha_k r_k^n$$

for $n = 0, 1, 2, \dots$, where $\alpha_1, \alpha_2, \dots, \alpha_k$ are constants.

23 The General Case with Repeated Roots Allowed

Theorem 4: Let $c_1, c_2, \dots, c_k$ be real numbers. Suppose that the characteristic equation

$$r^k - c_1 r^{k-1} - \dots - c_k = 0$$

has t distinct roots $r_1, r_2, \dots, r_t$ with multiplicities $m_1, m_2, \dots, m_t$, respectively so that $m_i \geq 1$ for $i = 1, 2, \dots, t$ and $m_1 + m_2 + \dots + m_t = k$. Then a sequence $\{a_n\}$ is a solution of the recurrence relation

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k}$$

if and only if

$$\begin{aligned} a_n = & (\alpha_{1,0} + \alpha_{1,1}n + \alpha_{1,2}n^2 + \dots + \alpha_{1,m_1-1}n^{m_1-1})r_1^n \\ & + (\alpha_{2,0} + \alpha_{2,1}n + \alpha_{2,2}n^2 + \dots + \alpha_{2,m_2-1}n^{m_2-1})r_2^n \\ & + \dots + (\alpha_{t,0} + \alpha_{t,1}n + \alpha_{t,2}n^2 + \dots + \alpha_{t,m_t-1}n^{m_t-1})r_t^n \end{aligned}$$

for $n = 0, 1, 2, \dots$, where $\alpha_{i,j}$ are constants for $1 \leq i \leq t$ and $0 \leq j \leq m_{i-1}$.

若 r 不只有一種解.....

<key>
 r^n

① $r = 2, 3$ $a_n = \alpha_1 2^n + \alpha_2 3^n$

nr^n

② $r = 2, 2$ $a_n = \alpha_1 2^n + \alpha_2 n 2^n$

$n^2 r^n$

③ $r = 2, 2, 2$ $a_n = \alpha_1 2^n + \alpha_2 n 2^n + \alpha_3 n^2 2^n$

$n^3 r^n$

④ $r = 2, 2, 3, 3, 3, 4, 4, 5, 5$

$\vdots$

$$a_n = \alpha_1 2^n + \alpha_2 n 2^n + \alpha_3 3^n + \alpha_4 n 3^n + \alpha_5 n^2 3^n + \alpha_6 4^n + \alpha_7 n 4^n + \alpha_8 5^n + \alpha_9 n 5^n$$

(數字第幾次
出現時)

24 Linear Nonhomogeneous Recurrence Relations with Constant Coefficients₁

線性非齊次問題

Definition: A **linear nonhomogeneous recurrence relation** with constant coefficients is a recurrence relation of the form:

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k} + F(n)$$

where $c_1, c_2, \dots, c_k$ are real numbers, and $F(n)$ is a function not identically zero depending only on n .

The recurrence relation

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k}$$

is called the **associated homogeneous recurrence relation**.

25 Linear Nonhomogeneous Recurrence Relations with Constant Coefficients₂

The following are **linear nonhomogeneous recurrence relations** with constant coefficients:

$$a_n = a_{n-1} + 2^n$$

$$a_n = a_{n-1} + a_{n-2} + n^2 + n + 1$$

$$a_n = 3a_{n-1} + n 3^n$$

$$a_n = a_{n-1} + a_{n-2} + a_{n-3} + n!$$

$$a_n = a_{n-1} + \underbrace{1}_{F(n)}$$

where the following are the **associated linear homogeneous recurrence relations**, respectively:

$$a_n = a_{n-1}$$

$$a_n = a_{n-1} + a_{n-2}$$

$$a_n = 3a_{n-1}$$

$$a_n = a_{n-1} + a_{n-2} + a_{n-3}$$

26 Solving Linear Nonhomogeneous Recurrence Relations with Constant Coefficients₁

Theorem 5: If $\{a_n^{(p)}\}$ is a particular solution of the **nonhomogeneous** linear recurrence relation with constant coefficients

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k} + F(n)$$

then every solution is of the form $\{a_n^{(p)} + a_n^{(h)}\}$, where $\{a_n^{(h)}\}$ is a solution of the associated homogeneous recurrence relation

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k}$$

27 Solving Linear Nonhomogeneous Recurrence Relations with Constant Coefficients₂

齊次解 (忽略 +2n) **Example:** Find all solutions of the recurrence relation $a_n = 3a_{n-1} + 2n$. What is the solution with $a_1 = 3$?

Solution: The associated linear homogeneous equation is $a_n = 3a_{n-1}$. (求出特解後)
↑ Its solutions are $a_n^{(h)} = \alpha 3^n$, where α is a constant. → 不能先把 α 算出來! 解定形後才能算 α !

↓ 特解 用猜的 Because $F(n) = 2n$ is a polynomial in n of degree one, to find a particular solution we might try a linear function in n , say $p_n = cn + d$, where c and d are constants. Suppose that $p_n = cn + d$ is such a solution. 目標 n 都要成立!

Then $a_n = 3a_{n-1} + 2n$ becomes $cn + d = 3(c(n-1) + d) + 2n$.

Simplifying yields $(2 + 2c)n + (2d - 3c) = 0$. It follows that $cn + d$ is a solution if and only if

$2 + 2c = 0$ and $2d - 3c = 0$. Therefore, $cn + d$ is a solution if and only if $c = -1$ and $d = -3/2$.

Consequently, $a_n^{(p)} = -n - 3/2$ is a particular solution.

By Theorem 5, all solutions are of the form $a_n = a_n^{(p)} + a_n^{(h)} = -n - 3/2 + \alpha 3^n$, where α is a constant.

To find the solution with $a_1 = 3$, let $n = 1$ in the above formula for the general solution.

Then $3 = -1 - 3/2 + 3\alpha$, and $\alpha = 11/6$. Hence, the solution is $a_n = -n - 3/2 + (11/6)3^n$.

Chapter 8.

Advanced Counting Techniques

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8.2 Solving Linear Recurrence Relations

8.3 Divide-and-Conquer Algorithms and Recurrence Relations

8.4 Generating Functions 分治法 遞迴方程

8.5 Inclusion-Exclusion

8.6 Applications of Inclusion-Exclusion

Divide-and-Conquer Algorithmic Paradigm

Definition: A *divide-and-conquer algorithm* works by first *dividing* a problem into one or more instances of **the same problem of smaller size** and then *conquering* the problem using the solutions of the smaller problems to find a solution of the original problem.

Examples:

- **Binary search**, covered in Chapters 3 and 5: It works by comparing the element to be located to the middle element. The original list is then split into two lists and the search continues recursively in the appropriate sublist.
- **Merge sort**, covered in Chapter 5: A list is split into two approximately equal sized sublists, each recursively sorted by merge sort. Sorting is done by successively merging pairs of lists.

Divide-and-Conquer Recurrence Relations

Suppose that a recursive algorithm divides a problem of size n into a subproblems.

Assume each subproblem is of size n/b .

Suppose $g(n)$ extra operations are needed in the conquer step.

Then $f(n)$ represents the number of operations to solve a problem of size n satisfies the following recurrence relation:

$$f(n) = af(n/b) + g(n)$$

This is called a ***divide-and-conquer recurrence relation***.

Example: Binary Search

Binary search reduces the search for an element in a sequence of size n to the search in a sequence of size $n/2$. Two comparisons are needed to implement this reduction;

- one to decide whether to search the upper or lower half of the sequence and
- the other to determine if the sequence has elements.

Hence, if $f(n)$ is the number of comparisons required to search for an element in a sequence of size n , then

$$f(n) = f(n/2) + 2$$

when n is even.

Example: Merge Sort

The merge sort algorithm splits a list of n (assuming n is even) items to be sorted into two lists with $n/2$ items. It uses fewer than n comparisons to merge the two sorted lists.

Hence, the number of comparisons required to sort a sequence of size n , is no more than $M(n)$ where

$$M(n) = 2M(n/2) + n.$$

Example: Fast Multiplication of Integers

快速幂

An algorithm for the fast multiplication of two $2n$ -bit integers (assuming n is even) first splits each of the $2n$ -bit integers into two blocks, each of n bits.

Suppose that a and b are integers with binary expansions of length $2n$. Let

$$a = (a_{2n-1}a_{2n-2} \dots a_1a_0)_2 \text{ and } b = (b_{2n-1}b_{2n-2} \dots b_1b_0)_2.$$

Let $a = 2^n A_1 + A_0$, $b = 2^n B_1 + B_0$, where

$$A_1 = (a_{2n-1} \dots a_{n+1}a_n)_2, A_0 = (a_{n-1} \dots a_1a_0)_2,$$

$$B_1 = (b_{2n-1} \dots b_{n+1}b_n)_2, B_0 = (b_{n-1} \dots b_1b_0)_2.$$

The algorithm is based on the fact that ab can be rewritten as:

$$ab = (2^{2n} + 2^n)A_1B_1 + 2^n(A_1 - A_0)(B_0 - B_1) + (2^n + 1)A_0B_0.$$

This identity shows that the multiplication of two $2n$ -bit integers can be carried out using three multiplications of n -bit integers, together with additions, subtractions, and shifts.

Hence, if $f(n)$ is the total number of operations needed to multiply two n -bit integers, then

$$f(2n) = 3f(n) + Cn$$

where Cn represents the total number of bit operations; the additions, subtractions and shifts that are a constant multiple of n -bit operations.

Estimating the Size of Divide-and-Conquer Functions₁

Theorem 1: Let f be an increasing function that satisfies the recurrence relation

$$f(n) = af\left(\frac{n}{b}\right) + c$$

whenever n is divisible by b , where $a \geq 1$, b is an integer greater than 1, and c is a positive real number. Then

$$f(n) \text{ is } \begin{cases} O(n^{\log_b a}) & \text{if } a > 1 \\ O(\log n) & \text{if } a = 1. \end{cases}$$

Furthermore, when $n = b^k$ and $a \neq 1$, where k is a positive integer,

$$f(n) = C_1 n^{\log_b a} + C_2$$

where $C_1 = f(1) + c / (a - 1)$ and $C_2 = -c / (a - 1)$.

$$f(n) = af\left(\frac{n}{b}\right) + C, \quad n \text{ 必须是 } b^k (n=b^k)$$

$$f(b^k) = af(b^{k-1}) + C$$

$$d_k = ad_{k-1} + \underbrace{C}_{\text{constant}}$$

< 齊次 solution >

$$d_k = ad_{k-1}$$

$$r^k = ar^{k-1}$$

$$r = a$$

$$d_k = \alpha_1 a^k$$

$$d_k = \alpha_1 a^k + \frac{C}{1-a}$$

< 特解 solution >

$$\text{guess: } d_k = p$$

, p is a constant

$$p = ap + C$$

$$p(1-a) = C$$

$$p = \frac{C}{1-a}$$

$$f(b^k) = d_k = \alpha_1 a^k + \frac{C}{1-a}$$

$$\begin{aligned} & \parallel \\ & f(n) \quad \log n = \log b^k \\ & \quad \quad \quad = k \log b \end{aligned}$$

$$\frac{\log n}{\log b} = k$$

$$\alpha_1 a^{\log_b n} + \frac{C}{1-a}$$

$$\underbrace{a^{\log_b n}}_{(\log_b)} = \underbrace{n^{\log_b a}}_{(\log_b)}$$

Explanation

Theorem 1: Let f be an increasing function that satisfies the recurrence relation

$$f(n) = af\left(\frac{n}{b}\right) + c$$

Furthermore, when $n = b^k$ and $a \neq 1$, where k is a positive integer,

$$f(n) = C_1 n^{\log_b a} + C_2$$

where $C_1 = f(1) + c / (a - 1)$ and $C_2 = -c / (a - 1)$.

$$f(b^k) = af(b^{k-1}) + c$$

可以想成

$$d_k = ad_{k-1} + c$$

Homogeneous solution:

$$f(n) = f(b^k) = d_k^{(h)} = C_1 a^k = C_1 a^{\log_b n} = C_1 n^{\log_b a}$$

Particular solution:

$$d_k^{(p)} = C_2$$

The Solution when there is a Repeated Root

Theorem 2: Let c_1 and c_2 be real numbers with $c_2 \neq 0$. Suppose that $r^2 - c_1 r - c_2 = 0$ has one **repeated root** r_0 . Then the sequence $\{a_n\}$ is a solution to the recurrence relation

$$a_n = c_1 a_{n-1} + c_2 a_{n-2}$$

if and only if

$$a_n = \alpha_1 r_0^n + \alpha_2 n r_0^n$$

for $n = 0, 1, 2, \dots$, where α_1 and α_2 are constants.

Complexity of Binary Search

Binary Search Example: Give a **big-O estimate** for the number of comparisons used by a binary search.

Solution: Since the number of comparisons used by binary search is $f(n) = f(n/2) + 2$ where n is even, by Theorem 1, it follows that $f(n)$ is **$O(\log n)$** .

34 Estimating the Size of Divide-and-Conquer Functions.

Theorem 1: Let f be an increasing function that satisfies the recurrence relation

$$f(n) = af\left(\frac{n}{b}\right) + c$$

whenever n is divisible by b , where $a \geq 1$, b is an integer greater than 1, and c is a positive real number. Then

$$f(n) \text{ is } \begin{cases} O(n^{\log_b a}) & \text{if } a > 1 \\ O(\log n) & \text{if } a = 1 \end{cases}$$

Furthermore, when $n = b^k$ and $a \neq 1$, where k is a positive integer,

$$f(n) = C_1 n^{\log_b a} + C_2$$

where $C_1 = f(1) + c/(a-1)$ and $C_2 = -c/(a-1)$.

Estimating the Size of Divide-and-conquer Functions₂

主定理

Theorem 2. Master Theorem: Let f be an increasing function that satisfies the recurrence relation

$$f(n) = af(n/b) + cn^d + C$$

whenever $n = b^k$, where k is a positive integer greater than 1, and c and d are real numbers with c positive and d nonnegative. Then

$$f(n) \text{ is } \begin{cases} O(n^d) & \text{if } a < b^d, \\ O(n^d \log n) & \text{if } a = b^d, \\ O(n^{\log_b a}) & \text{if } a > b^d. \end{cases}$$

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Divide And Conquer:

P. 3-4

Complexity of Merge Sort

Merge Sort Example: Give a **big-O estimate** for the number of comparisons used by merge sort.

Solution: Since the number of comparisons used by merge sort to sort a list of n elements is less than $M(n)$ where $M(n) = 2M(n/2) + n$, by the master theorem $M(n)$ is **$O(n \log n)$** .

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Estimating the Size of Divide-and-conquer Functions.

Theorem 2. Master Theorem: Let f be an increasing function that satisfies the recurrence relation

$$f(n) = af(n/b) + cn^d$$

whenever $n = b^k$, where k is a positive integer greater than 1, and c and d are real numbers with c positive and d nonnegative. Then

$$f(n) \text{ is } \begin{cases} O(n^d) & \text{if } a < b^d, \\ O(n^d \log n) & \text{if } a = b^d, \\ O(n^{\log_b a}) & \text{if } a > b^d. \end{cases}$$

Complexity of Fast Integer Multiplication Algorithm

Integer Multiplication Example: Give a **big- O estimate** for the number of bit operations used needed to multiply two n -bit integers using the fast multiplication algorithm.

Solution: We have shown that $f(n) = 3f(n/2) + Cn$, when n is even, where $f(n)$ is the number of bit operations needed to multiply two n -bit integers. Hence by the master theorem with $a = 3$, $b = 2$, $c = C$, and $d = 0$ (so that we have the case where $a > b^d$), it follows that $f(n)$ is $O(n^{\log_2 3})$.

快速冪時間複雜度

37 Estimating the Size of Divide-and-conquer Functions:

Theorem 2. Master Theorem: Let f be an increasing function that satisfies the recurrence relation

$$f(n) = af(n/b) + cn^d$$

whenever $n = b^k$, where k is a positive integer greater than 1, and c and d are real numbers with c positive and d nonnegative. Then

$$f(n) \text{ is } \begin{cases} O(n^d) & \text{if } a < b^d, \\ O(n^d \log n) & \text{if } a = b^d, \\ O(n^{\log_b a}) & \text{if } a > b^d \end{cases}$$

Chapter 8.

Advanced Counting Techniques

8.1 Applications of Recurrence Relations

8.2 Solving Linear Recurrence Relations

8.3 Divide-and-Conquer Algorithms and Recurrence Relations

8.4 Generating Functions

8.5 Inclusion-Exclusion

8.6 Applications of Inclusion-Exclusion

Generating Functions

Definition: The **generating function (生成函數)** for the sequence $a_0, a_1, \dots, a_k, \dots$ of real numbers is the infinite series

$$G(x) = a_0 + a_1x + \dots + a_kx^k + \dots = \sum_{k=0}^{\infty} a_kx^k$$

Examples:

* 有一連串可能無窮的數列(係數), 把他改寫成 $\sum_{k=0}^{\infty} a_kx^k$ 形式

- The sequence $\{a_k\}$ with $a_k = 3$ has the generating function $\sum_{k=0}^{\infty} 3x^k$.
3, 3, 3, 3, 3, ...
- The sequence $\{a_k\}$ with $a_k = k + 1$ has the generating function $\sum_{k=0}^{\infty} (k + 1)x^k$.
1, 2, 3, 4, 5, ...
- The sequence $\{a_k\}$ with $a_k = 2^k$ has the generating function $\sum_{k=0}^{\infty} 2^k x^k$.
1, 2, 4, 8, 16, ...

Generating Functions for Finite Sequences₁

Generating functions for finite sequences of real numbers can be defined by extending a finite sequence $a_0, a_1, \dots, a_n$ into an infinite sequence by setting $a_{n+1} = 0$, $a_{n+2} = 0$, and so on.

有限的数列 ($a_0 \sim a_n$)
Assume a_{n+1} 项开始是 0

The generating function $G(x)$ of this infinite sequence $\{a_n\}$ is a **polynomial of degree n** because no terms of the form $a_j x^j$ with $j > n$ occur, that is,

$$G(x) = a_0 + a_1 x + \cdots + a_n x^n.$$

Generating Functions for Finite Sequences₂

Example: What is the generating function for the sequence **1, 1, 1, 1, 1, 1**?

Solution: The generating function of 1, 1, 1, 1, 1, 1 is

$$1 + x + x^2 + x^3 + x^4 + x^5. \quad , x=1 \text{ 時 } , = 6$$

By Theorem 1 of Section 2.4, we have $(x^6 - 1)/(x - 1) = 1 + x + x^2 + x^3 + x^4 + x^5$ $\left\{ \begin{array}{l} \frac{x^6 - 1}{x - 1} , x \neq 1 \\ 6 , x = 1 \end{array} \right.$ $\sim x \text{ 不能等於 } 1$

when $x \neq 1$.

Consequently **$G(x) = (x^6 - 1)/(x - 1)$** is the generating function of the sequence.

Useful Generating Functions₁

TABLE 1 Useful Generating Functions.

$G(x)$	a_k
$(1+x)^n = \sum_{k=0}^n C(n, k)x^k$ $= 1 + C(n, 1)x + C(n, 2)x^2 + \dots + x^n$	$C(n, k)$
$(1+ax)^n = \sum_{k=0}^n C(n, k)a^k x^k$ $= 1 + C(n, 1)ax + C(n, 2)a^2x^2 + \dots + a^n x^n$	$C(n, k)a^k$
$(1+x^r)^n = \sum_{k=0}^n C(n, k)x^{rk}$ $= 1 + C(n, 1)x^r + C(n, 2)x^{2r} + \dots + x^{rn}$	$C(n, k/r)$ if $r \mid k$; 0 otherwise
$\frac{1-x^{n+1}}{1-x} = \sum_{k=0}^n x^k = 1 + x + x^2 + \dots + x^n$	1 if $k \leq n$; 0 otherwise
$\frac{1}{1-x} = \sum_{k=0}^{\infty} x^k = 1 + x + x^2 + \dots$	1
$\frac{1}{1-ax} = \sum_{k=0}^{\infty} a^k x^k = 1 + ax + a^2x^2 + \dots$	a^k

$x=1$, Ans 為 2 的冪次

$x=-1$, Ans = 0

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級數

Useful Generating Functions₂

$\frac{1}{1-x^r} = \sum_{k=0}^{\infty} x^{rk} = 1 + x^r + x^{2r} + \dots$	1 if $r \mid k$; 0 otherwise
$\frac{1}{(1-x)^2} = \sum_{k=0}^{\infty} (k+1)x^k = 1 + 2x + 3x^2 + \dots$	$k+1$
$\begin{aligned} \frac{1}{(1-x)^n} &= \sum_{k=0}^{\infty} C(n+k-1, k)x^k \\ &= 1 + C(n, 1)x + C(n+1, 2)x^2 + \dots \end{aligned}$	$C(n+k-1, k) = C(n+k-1, n-1)$
$\begin{aligned} \frac{1}{(1+x)^n} &= \sum_{k=0}^{\infty} C(n+k-1, k)(-1)^k x^k \\ &= 1 - C(n, 1)x + C(n+1, 2)x^2 - \dots \end{aligned}$	$(-1)^k C(n+k-1, k) = (-1)^k C(n+k-1, n-1)$
$\begin{aligned} \frac{1}{(1-ax)^n} &= \sum_{k=0}^{\infty} C(n+k-1, k)a^k x^k \\ &= 1 + C(n, 1)ax + C(n+1, 2)a^2 x^2 + \dots \end{aligned}$	$C(n+k-1, k)a^k = C(n+k-1, n-1)a^k$
$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$	$1/k!$
$\ln(1+x) = \sum_{k=0}^{\infty} \frac{(-1)^{k+1}}{k} x^k = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$	$(-1)^{k+1}/k$

Counting Problems and Generating Functions₁

Example: Find the number of solutions of

$$e_1 + e_2 + e_3 = 17,$$

where e_1, e_2 , and e_3 are nonnegative integers with $2 \leq e_1 \leq 5, 3 \leq e_2 \leq 6$, and $4 \leq e_3 \leq 7$.

x^{17} 的係數

Solution: The number of solutions is the **coefficient of x^{17}** in the expansion of

$$(x^2 + x^3 + x^4 + x^5) (x^3 + x^4 + x^5 + x^6) (x^4 + x^5 + x^6 + x^7).$$

This follows because a term equal to x^{17} is obtained in the product by picking a term in the first sum x^{e_1} , a term in the second sum x^{e_2} , and a term in the third sum x^{e_3} , where $e_1 + e_2 + e_3 = 17$.

There are three solutions since the coefficient of x^{17} in the product is 3.

Counting Problems and Generating Functions₂

Example: Use generating functions to find the number of k -combinations of a set with n elements, i.e., $C(n,k)$.

Solution: Each of the n elements in the set contributes the term $(1 + x)$ to the generating function

$$f(x) = \sum_{k=0}^n a^k x^k.$$

Hence $f(x) = (1 + x)^n$ where $f(x)$ is the generating function for $\{a^k\}$, where a^k represents the number of k -combinations of a set with n elements.

By the binomial theorem, we have

$$f(x) = \sum_{k=0}^n \binom{n}{k} x^k$$

where $\binom{n}{k} = \frac{n!}{k!(n-k)!}$

Hence, $C(n,k) = \frac{n!}{k!(n-k)!}$

TABLE 1 Useful Generating Functions.	
$G(x)$	a_k
$(1+x)^n = \sum_{k=0}^n C(n,k)x^k$ $= 1 + C(n,1)x + C(n,2)x^2 + \cdots + x^n$	$C(n,k)$

8.4 Generating Functions

(補充)

11

Counting the Ways to Parenthesize a Product 8.2

Example: Find a recurrence relation for C_n , the number of ways to parenthesize the product of $n + 1$ numbers, $x_0 \cdot x_1 \cdot x_2 \cdot \dots \cdot x_n$, to specify the order of multiplication. For example, $C_3 = 5$, since all the possible ways to parenthesize 4 numbers are

$$\begin{aligned} &((x_0 \cdot x_1) \cdot x_2) \cdot x_3 & (x_0 \cdot (x_1 \cdot x_2)) \cdot x_3 & (x_0 \cdot x_1) \cdot (x_2 \cdot x_3) \\ &x_0 \cdot ((x_1 \cdot x_2) \cdot x_3) & x_0 \cdot (x_1 \cdot (x_2 \cdot x_3)) \end{aligned}$$

Solution: Note that however parentheses are inserted in $x_0 \cdot x_1 \cdot x_2 \cdot \dots \cdot x_n$, one “.” operator remains outside all parentheses. This final operator appears between two of the $n + 1$ numbers, say x_k and x_{k+1} . Since there are C_k ways to insert parentheses in the product $x_0 \cdot x_1 \cdot x_2 \cdot \dots \cdot x_k$ and C_{n-k-1} ways to insert parentheses in the product $x_{k+1} \cdot x_{k+2} \cdot \dots \cdot x_n$, we have

$$\begin{aligned} C_n &= C_0 C_{n-1} + C_1 C_{n-2} + \dots + C_{n-2} C_1 + C_{n-1} C_0 \\ &= \sum_{k=0}^{n-1} C_k C_{n-k-1} \end{aligned}$$

The initial conditions are $C_0 = 1$ and $C_1 = 1$.

The sequence $\{C_n\}$ is the sequence of **Catalan Numbers**. This recurrence relation can be solved using the method of generating functions; see Exercise 41 in Section 8.4.

How to compute C_n ?

How to compute C_n ?

- Trick: Assume we let $C(x) = \sum_{i \geq 0} C_i x^i$ which is the generating function for the number of binary trees.

$$C(x) = \underbrace{C_0}_{=1} + C_1 x + C_2 x^2 + C_3 x^3 + C_4 x^4 + \dots$$

$$C_n = C_0 C_{n-1} + C_1 C_{n-2} + \dots + C_{n-2} C_1 + C_{n-1} C_0$$

$$= \sum_{k=0}^{n-1} C_k C_{n-k-1} \quad (C_0 = 1 \text{ and } C_1 = 1)$$

$$C^2(x) = C_0 C_0 + (C_0 C_1 + C_1 C_0)x + (C_0 C_2 + C_1 C_1 + C_2 C_0)x^2 + (C_0 C_3 + C_1 C_2 + C_2 C_1 + C_3 C_0)x^3 + \dots$$

都缺了一个 x 为

$$C(x)C(x) = (C_0 + C_1 x + C_2 x^2 + \dots)(C_0 + C_1 x + C_2 x^2 + \dots)$$

- By the recurrence relation we get

$$xC^2(x) = C(x) - 1$$

How to compute C_n ?

- Trick: Assume we let $C(x) = \sum_{i \geq 0} C_i x^i$ which is the generating function for the number of binary trees.

- By the recurrence relation we get

$$xC^2(x) = C(x) - 1$$

$$xC^2(x) - C(x) + 1 = 0$$

$$C(x) = \frac{1 - \sqrt{1 - 4x}}{2x} \rightarrow (1 - 4x)^{\frac{1}{2}} = (1 + (-4x))^{\frac{1}{2}}$$

$$(-4x)^n = (-1)^n 2^{2n} x^n$$

廣義二項式定理

$$C(x) = \frac{1}{2x} \left(1 - \sum_{n \geq 0} \binom{1/2}{n} (-4x)^n \right) = \sum_{m \geq 0} \binom{1/2}{m+1} \frac{(-1)^m 2^{2m+1} x^m}{x}$$

$$C_n = \binom{1/2}{n+1} (-1)^n 2^{2n+1} = \sum_{n \geq 1} \binom{1/2}{n} (-1)^{n+1} 2^{2n-1} x^{n-1} \quad \text{令 } n=m+1$$

$$C_n = \binom{1/2}{n+1} (-1)^n 2^{2n+1}$$

$$= \frac{\frac{1}{2} \left(\frac{1}{2}-1\right) \left(\frac{1}{2}-2\right) \cdots \left(\frac{1}{2}-n\right)}{(n+1)(n)(n-1)(n-2) \cdots 1} (-1)^n 2^{2n+1}$$

$$= \frac{\frac{1}{2} \left(1-\frac{1}{2}\right) \left(2-\frac{1}{2}\right) \cdots \left(n-\frac{1}{2}\right)}{(n+1)(n)(n-1)(n-2) \cdots 1} 2^{2n+1}$$

$$= \left(\frac{1}{2}\right)^{n+1} \frac{1(2-1)(4-1) \cdots (2n-1)}{(n+1)(n)(n-1)(n-2) \cdots 1} 2^{2n+1}$$

$$= \frac{1(2-1)(4-1) \cdots (2n-1)}{(n+1)(n)(n-1)(n-2) \cdots 1} 2^n$$

$$= \frac{1(1)(3) \cdots (2n-1)}{(n+1)(n)(n-1)(n-2) \cdots 1} 2^n$$

$$= \frac{1(1)(3) \cdots (2n-1)}{(n+1)(n)(n-1)(n-2) \cdots 1} 2^n \frac{2 \cdot 4 \cdot 6 \cdot 8 \cdots (2n)}{2 \cdot 4 \cdot 6 \cdot 8 \cdots (2n)}$$

$$= \frac{1(1)(3) \cdots (2n-1)}{(n+1)(n)(n-1)(n-2) \cdots 1} 2^n \frac{2 \cdot 4 \cdot 6 \cdot 8 \cdots (2n)}{2^n \cdot 1 \cdot 2 \cdot 3 \cdots (n)} \quad n!$$

$$= \frac{(2n)!}{(n+1)(n!)} \frac{1}{2^n \cdot (n!)} = \frac{1}{n+1} \binom{2n}{n}$$

$$= \frac{(2n)!}{(n+1)(n!)} = \binom{2n}{n} \times \frac{1}{n+1}$$

$$\binom{10}{3} = \frac{10 \cdot 9 \cdot 8}{3 \cdot 2 \cdot 1} = \frac{10 \cdot 9 \cdot 8}{3!}$$

Example.

$$C_1 = \frac{1}{2} \binom{2}{1} = \frac{2}{2} = 1$$

$$C_2 = \frac{1}{3} \binom{4}{2} = \frac{6}{3} = 2$$

$$C_3 = \frac{1}{4} \binom{6}{3} = \frac{20}{4} = 5$$

11 Counting the Ways to Parenthesize a Product

Example: Find a recurrence relation for C_n , the number of ways to parenthesize the product of $n+1$ numbers, $x_0 \cdot x_1 \cdot x_2 \cdots x_n$, to specify the order of multiplication. For example, $C_3 = 5$, since all the possible ways to parenthesize 4 numbers are

$$\begin{array}{ccc} (x_0 \cdot x_1) \cdot x_2 \cdot x_3 & (x_0 \cdot (x_1 \cdot x_2)) \cdot x_3 & (x_0 \cdot x_2) \cdot (x_1 \cdot x_3) \\ x_0 \cdot ((x_1 \cdot x_2) \cdot x_3) & x_0 \cdot (x_1 \cdot (x_2 \cdot x_3)) & \end{array}$$

Chapter 8.

Advanced Counting Techniques

8.1 Applications of Recurrence Relations

8.2 Solving Linear Recurrence Relations

8.3 Divide-and-Conquer Algorithms and Recurrence Relations

8.4 Generating Functions

8.5 Inclusion-Exclusion 排容原理

8.6 Applications of Inclusion-Exclusion

Principle of Inclusion-Exclusion

(排容原理)

In Section 2.2, we developed the following formula for the number of elements in the union of two finite sets:

$$|A \cup B| = |A| + |B| - |A \cap B|$$

We will generalize this formula to finite sets of any size.

<key>
counting < Ch.6
Ch.8

Two Finite Sets

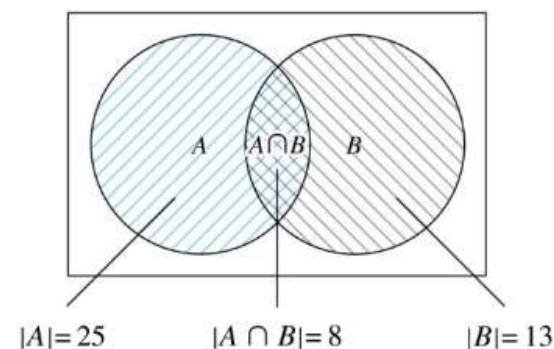
Example: In a discrete mathematics class every student is a major in computer science or mathematics or both. The number of students having computer science as a major (possibly along with mathematics) is **25**; the number of students having mathematics as a major (possibly along with computer science) is **13**; and the number of students majoring in both computer science and mathematics is **8**. How many students are in the class?

Solution:

$$|A \cup B| = |A| + |B| - |A \cap B|$$

$$= 25 + 13 - 8 = 30$$

$$|A \cup B| = |A| + |B| - |A \cap B| = 25 + 13 - 8 = 30$$

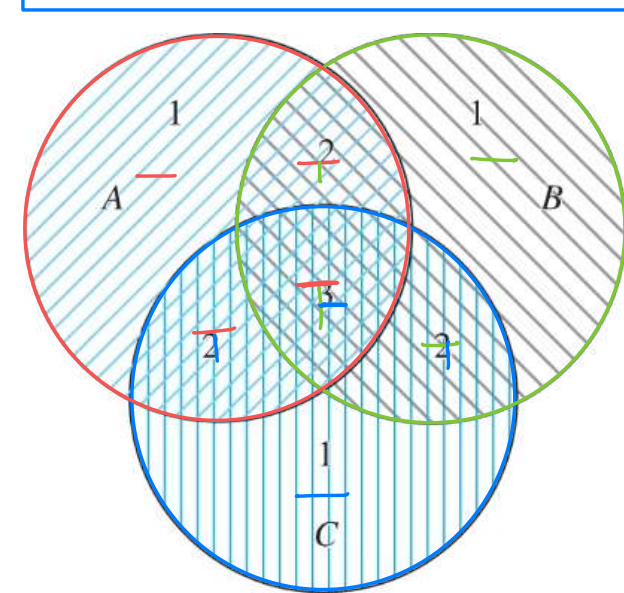


[Jump to long description](#)

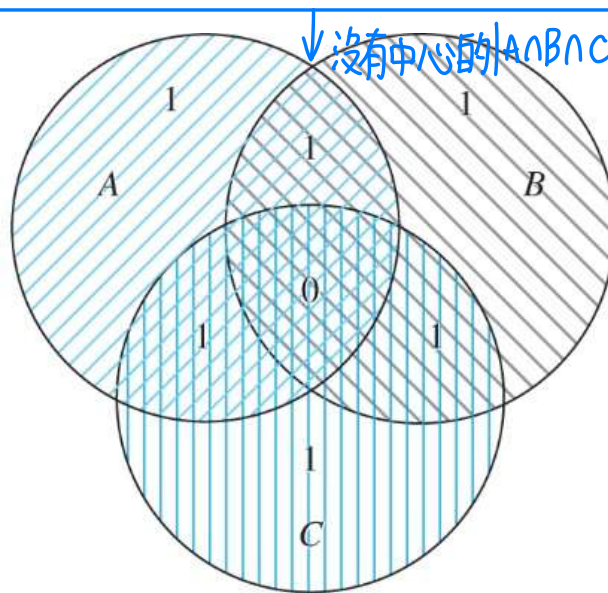
Three Finite Sets₁

$$|A \cup B \cup C| =$$

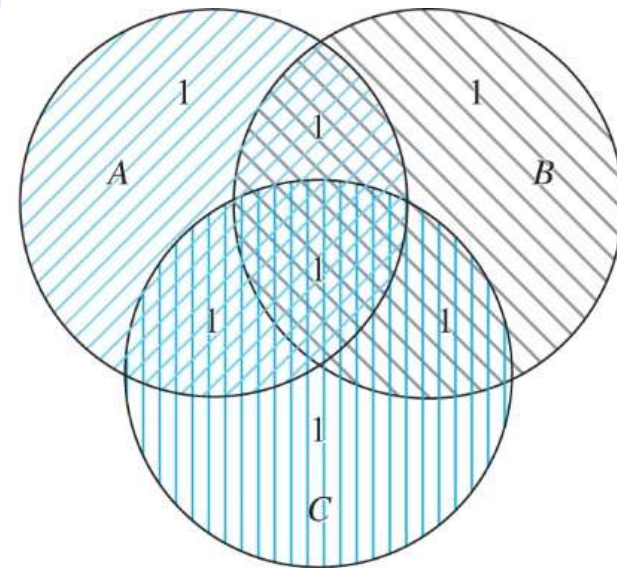
$$|A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$



(a) Count of elements by
 $|A| + |B| + |C|$



(b) Count of elements by
 $|A| + |B| + |C| - |A \cap B| -$
 $|A \cap C| - |B \cap C|$

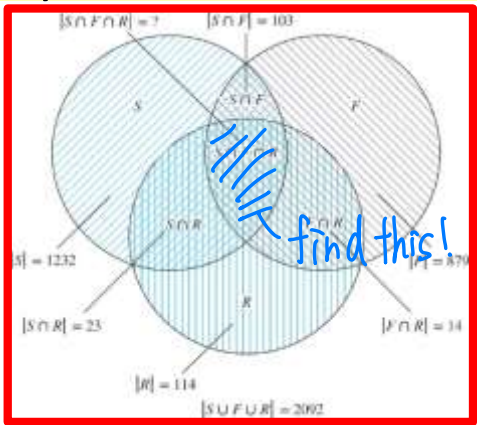


(c) Count of elements by
 $|A| + |B| + |C| - |A \cap B| -$
 $|A \cap C| - |B \cap C| + |A \cap B \cap C|$

Three Finite Sets₂

Example: A total of 1232 students have taken a course in S, 879 have taken a course in F, and 114 have taken a course in R. Further, 103 have taken courses in both S and F, 23 have taken courses in both S and R, and 14 have taken courses in both F and R. If 2092 students have taken a course in at least one of S, F, and R, how many students have taken a course in all 3?

Solution:



Ch6. 圖+小問題
Counting? < Ch8. 魔回

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$

$|S| = 1232$, $|F| = 879$, $|R| = 114$, $|S \cap F| = 103$, $|S \cap R| = 23$, $|F \cap R| = 14$, and $|S \cup F \cup R| = 2092$.

Using the equation

$$|S \cup F \cup R| = |S| + |F| + |R| - |S \cap F| - |S \cap R| - |F \cap R| + |S \cap F \cap R|,$$

we obtain $2092 = 1232 + 879 + 114 - 103 - 23 - 14 + |S \cap F \cap R|$.

Solving for $|S \cap F \cap R|$ yields 7.

The Principle of Inclusion-Exclusion₁

Theorem 1. The Principle of Inclusion-Exclusion:

Let $A_1, A_2, \dots, A_n$ be finite sets. Then:

$$|A_1 \cup A_2 \cup \dots \cup A_n| = \sum_{1 \leq i \leq j \leq k \leq n} |A_i| - \sum_{1 \leq i \leq j \leq n} |A_i \cap A_j| + \sum_{1 \leq i \leq j \leq k \leq n} |A_i \cap A_j \cap A_k| - \dots + (-1)^{n+1} |A_1 \cap A_2 \cap \dots \cap A_n|$$

The Principle of Inclusion-Exclusion₂

Proof: An element in the union is counted exactly once in the right-hand side of the equation. Consider an element a that is a member of r of the sets $A_1, \dots, A_n$ where $1 \leq r \leq n$.

- It is counted $C(r,1)$ times by $\sum |A_i|$
- It is counted $C(r,2)$ times by $\sum |A_i \cap A_j|$
- In general, it is counted $C(r,m)$ times by the summation of m of the sets A_i .

The Principle of Inclusion-Exclusion₃

Thus the element is counted **exactly**

$$C(r,1) - C(r,2) + C(r,3) - \cdots + (-1)^{r+1} C(r,r) = \mathbf{1}.$$

times by the right hand side of the equation.

Proof.

By Corollary 2 of Section 6.4, we have

$$C(r,0) - C(r,1) + C(r,2) - \cdots + (-1)^r C(r,r) = 0.$$

Hence,

$$1 = C(r,0) = C(r,1) - C(r,2) + \cdots + (-1)^{r+1} C(r,r).$$

Chapter 8.

Advanced Counting Techniques

8.1 Applications of Recurrence Relations

8.2 Solving Linear Recurrence Relations

8.3 Divide-and-Conquer Algorithms and Recurrence Relations

8.4 Generating Functions

8.5 Inclusion-Exclusion

8.6 Applications of Inclusion-Exclusion

The Number of Onto Functions¹

(覆盖所有值域 → 画图! 

Example: How many onto functions are there from a set with six elements to a set with three elements?

Solution: Suppose that the elements in the codomain are b_1, b_2 , and b_3 . Let P_1, P_2 , and P_3 be the properties that b_1, b_2 , and b_3 are not in the range of the function, respectively. The function is onto if none of the properties P_1, P_2 , and P_3 hold.

By the inclusion-exclusion principle the number of onto functions from a set with six elements to a set with three elements is

$$N - [N(P_1) + N(P_2) + N(P_3)] + [N(P_1P_2) + N(P_1P_3) + N(P_2P_3)] - N(P_1P_2P_3)$$

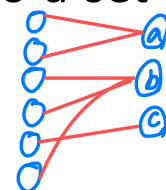
- Here the total number of functions from a set with six elements to one with three elements is $N = 3^6$.
- The number of functions that do not have in the range is $N(P_1) = 2^6$. Similarly, $N(P_2) = N(P_3) = 2^6$.
- Note that $N(P_1P_2) = N(P_1P_3) = N(P_2P_3) = 1$ and $N(P_1P_2P_3) = 0$.

<key> 交集是否容易计算

<key>: $C_6^3 - C_1^3$

Hence, the number of onto functions from a set with six elements to a set with three elements is:

$$3^6 - 3 \cdot 2^6 + 3 = 729 - 192 + 3 = 540$$



The Number of Onto Functions₂

Theorem 1: Let m and n be positive integers with $m \geq n$. Then there are

$$n^m - C(n, 1)(n-1)^m + C(n, 2)(n-2)^m - \cdots + (-1)^{n-1} C(n, n-1) \cdot 1^m$$

onto functions from a set with m elements to a set with n elements.

Proof follows from the principle of inclusion-exclusion (see *Exercise 27*).

Derangements₁

find 幾個錯位!

Definition: A **derangement** (錯位) is a permutation of objects that leaves **no object in the original position**.

like: 原始 1, 2, 3, 4, 5
錯位 2, 1, 5, 3, 4

Example: The permutation of 21453 is a derangement of 12345 because no number is left in its original position. But 21543 is not a derangement of 12345, because 4 is in its original position. <從對位開始排, 扣掉!> (排組畫圖)

$$5! - C_1^5 4! + C_2^5 3! - C_3^5 2! + C_4^5 1! - C_5^5 0!$$

Derangements₂

Theorem 2: The number of derangements of a set with n elements is

$$D_n = n! \left[1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \cdots + (-1)^n \frac{1}{n!} \right].$$

Example.

The derangements of $\{1,2,3\}$ are $\{2,3,1\}$ and $\{3,1,2\}$.

The number derangements of a 3-element set is

$$3! \cdot \sum_{k=0}^3 \frac{(-1)^k}{k!} = 6 \cdot \left(\frac{1}{1} - \frac{1}{1} + \frac{1}{2} - \frac{1}{6} \right) = 2$$